

Test 13 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Use the drop-down menu to complete the statistical question.

How many minutes did [[1]] spend reading last night?

Blank (1) choices: each student in the class | I | my teacher

Show your work

editing task choice

2. Select **all** the expressions equivalent to $6(x + 8)$.

- ☐ $6x + 8$
- ☐ $48 + 6x$
- ☐ $14(x)$
- ☐ $6(x + 8)$
- ☐ $6x + 48$
- ☐ $6(x + 9)$

Select all that apply.

Show your work

multiselect

3. Select **all** the numbers whose absolute value is 9.

Circle every correct choice:

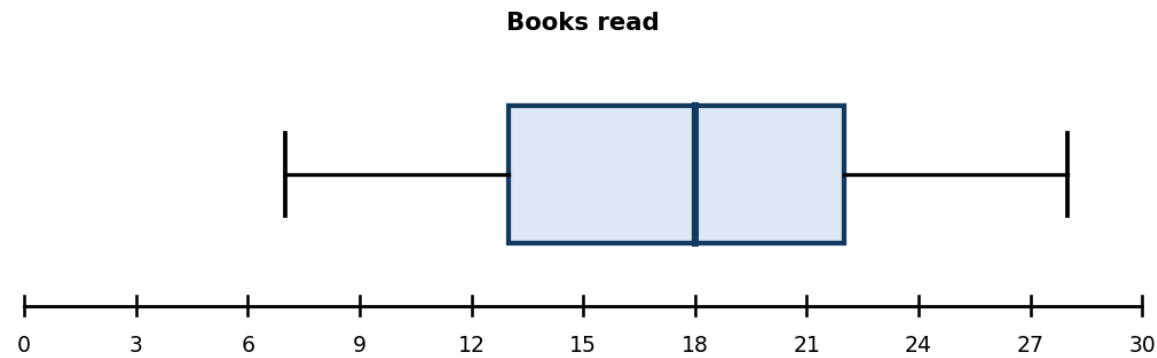
- ☐ 9
- ☐ -9
- ☐ 18
- ☐ -18
- ☐ 0
- ☐ 10

Show your work

calculator · selectable hot text

4. The box plot shows a set of data.

Select **all** the values that lie between the lower quartile and the upper quartile, inclusive.



- ☐ 31
- ☐ 13
- ☐ 22
- ☐ 28
- ☐ 7
- ☐ 18
- ☐ 15

Select all that apply.

Show your work

multiselect

5. A teacher has 9 pencils and 15 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: _____

Show your work

calculator · equation editor

6. Select **all** the ratios equivalent to 8:14.

- ☐ 4:7
- ☐ 5:7
- ☐ 7:4
- ☐ 16:28
- ☐ 12:21
- ☐ 4:9

Select all that apply.

Show your work

multiselect

7. Select **all** the ratios that belong in a table of ratios equivalent to 5:6.

- ☐ 10:12
- ☐ 5:6
- ☐ 6:5
- ☐ 6:7
- ☐ 20:24
- ☐ 10:6

Select all that apply.

Show your work

multiselect

8. Select **all** the expressions with a value of 22 when $x = 6$.

- ☐ $10 + 2x$
- ☐ $2x + 10$
- ☐ $2(x + 10)$
- ☐ $2x - 10$
- ☐ $12x$
- ☐ $2(x) + 10$

Select all that apply.

Show your work

multiselect

9. A right rectangular prism measures 4 by 5 by 2 inches.

Select **all** the expressions that give its volume in cubic inches.

- ☐ $4 \times 5 \times 2$
- ☐ $2 \times 5 \times 4$
- ☐ 20×2
- ☐ $2 \times (4 + 5 + 2)$
- ☐ 4×5
- ☐ $4 + 5 + 2$

Select all that apply.

Show your work

multiselect

10. Select **all** the points that lie on the vertical line through (3, 0).

- ☐ (3, 6)
- ☐ (5, -5)
- ☐ (3, 2)
- ☐ (5, 1)
- ☐ (0, 4)
- ☐ (3, -3)

Select all that apply.

Show your work

multiselect

11. A data set is shown.

12, 12, 13, 17, 17, 19

The value 29 is added to the data set.

Select **all** the measures that change.

- ☐ Range
- ☐ Maximum
- ☐ Median
- ☐ Mean
- ☐ Minimum

Select all that apply.

Show your work

multiselect

12. Select **all** the equations that have the solution $x = 15$.

☐ $x + 8 = 23$

☐ $x + 15 = 8$

☐ $x + 8 = 15$

☐ $8 + x = 23$

☐ $x - 8 = 7$

☐ $x - 8 = 15$

Select all that apply.

Show your work

multiselect

13. A triangle has a base of 8 centimeters and a height of 9 centimeters.

Select **all** the expressions that give its area in square centimeters.

☐ $0.5 \times 8 \times 9$

☐ $(8 \times 9) \div 2$

☐ 8×9

☐ $8 \times 9 \times 0.5$

☐ $(8 + 9) \div 2$

☐ $2 \times 8 \times 9$

Select all that apply.

Show your work

multiselect

14. On a January morning the temperature in Homestead was 12 degrees below zero Celsius at a research freezer.

That temperature is °C, and its opposite is °C.

Blank (1) choices: -12 | 12 | -24 | 0

Blank (2) choices: -12 | 12 | -24 | 0

Show your work

calculator · editing task choice

15. Select **all** the values of x that make $x \leq 9$ true.

- ☐ 11
- ☐ 6
- ☐ 14
- ☐ 9
- ☐ 8
- ☐ 17

Select all that apply.

Show your work

multiselect

16. Select **all** the equations that have the solution $x = 6$.

- ☐ $7x = 42$
- ☐ $x + 7 = 6$
- ☐ $7x = 6$
- ☐ $42 = 7x$
- ☐ $x \div 7 = 42$
- ☐ $x \div 7 = 6/7$

Select all that apply.

Show your work

multiselect

17. A data set is shown.

1, 3, 5, 5, 9, 9, 10, 13, 17, 19, 22, 23, 23, 23, 24, 25, 28, 29

Select **all** the histogram intervals that would contain exactly 2 values.

- ☐ 10-14
- ☐ 25-29
- ☐ 15-19
- ☐ 0-4
- ☐ 20-24
- ☐ 5-9

Select all that apply.

Show your work

multiselect

18. What is the greatest common factor you would use to rewrite $20 + 8$ as a product? Enter your answer.

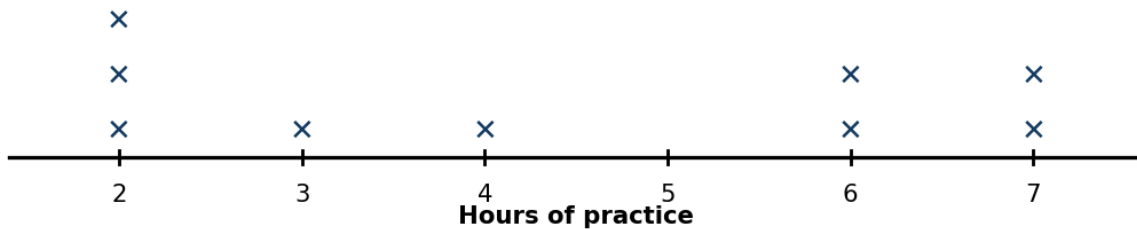
Answer: _____

Show your work

calculator · equation editor

19. The line plot shows hours of practice.

Select **all** the values that were reported by exactly 2 students.



- ☐ 7
- ☐ 2
- ☐ 4
- ☐ 3
- ☐ 6

Select all that apply.

Show your work

multiselect

20. Enter the prime factorization of 50. Use ^ for exponents and $\times$ between factors, for example $2^2 \times 3$.

Answer: _____

Show your work

calculator · equation editor

21. A roll holds 3 yards of ribbon. Ana cuts 8 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

22. Use the drop-down menu to make the statement true.

$|-19|$ $[[1]]$ $|7|$

Blank (1) choices: $<$ $|$ $>$ $=$

Show your work

calculator · editing task choice

23. What is the value of $30 + (-2)$?

- A. 28
- B. -28
- C. 30
- D. 32

Show your work

calculator · multiple choice

24. Marcos needs $4\frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 4 batches? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

25. A right rectangular prism measures 5 by 5 by 2 centimeters.

Select **all** the expressions that give its surface area in square centimeters.

- ☐ 90×1
- ☐ $2(5 \times 5 + 5 \times 2 + 5 \times 2)$
- ☐ $2(5 \times 5) + 2(5 \times 2) + 2(5 \times 2)$
- ☐ $6 \times 5 \times 5$
- ☐ $5 \times 5 + 5 \times 2 + 5 \times 2$
- ☐ $5 \times 5 \times 2$

Select all that apply.

Show your work

multiselect

26. A parallelogram has a base of 8 centimeters and a height of 4 centimeters.

Select **all** the expressions that give its area in square centimeters.

- ☐ $2 \times (8 + 4)$
- ☐ 32×1
- ☐ 4×8
- ☐ $8 + 4$
- ☐ 8×4
- ☐ $(8 \times 4) \div 2$

Select all that apply.

Show your work

multiselect

27. Select **all** the expressions equivalent to 6 more than 4 times a number x .

- ☐ $4x - 6$
- ☐ $4x + 6$
- ☐ $10x$
- ☐ $x \cdot 4 + 6$
- ☐ $6 + 4x$
- ☐ $4(x + 6)$

Select all that apply.

Show your work

multiselect

28. What is the value of 0.3^2 ? Enter your answer.

Answer: _____

Show your work

calculator · equation editor

29. What is the value of $(-9)(9)$?

- A. -77
- B. 0
- C. 81
- D. -81

Show your work

calculator · multiple choice

30. A 8-pound box costs \$64.00.

Select **all** the expressions that give the cost per pound, in dollars.

- ☐ $64 \div 8$
- ☐ $8 \div 64$
- ☐ $64 - 8$
- ☐ 64×8
- ☐ $64/8$
- ☐ $128 \div 16$

Select all that apply.

Show your work

multiselect

31. A store sells plantains for \$1.93 each.

What is the total cost of 16 of them? Enter your answer in dollars.

Answer: _____

Show your work

calculator · equation editor

32. Select **all** the expressions that give 15% of 80.

- ☐ $(15/100) \times 80$
- ☐ 80×0.15
- ☐ $15\% \times 80$
- ☐ $80 - 15$
- ☐ $80 \div 15$
- ☐ 15×80

Select all that apply.

Show your work

multiselect

33. Use the drop-down menus to place these numbers in order from least to greatest.

-4.5, $-1 \frac{3}{4}$, $\frac{2}{5}$, 1

Least , , , greatest.

Blank (1) choices: -4.5 | $-1 \frac{3}{4}$ | $\frac{2}{5}$ | 1

Blank (2) choices: -4.5 | $-1 \frac{3}{4}$ | $\frac{2}{5}$ | 1

Blank (3) choices: -4.5 | $-1 \frac{3}{4}$ | $\frac{2}{5}$ | 1

Blank (4) choices: -4.5 | $-1 \frac{3}{4}$ | $\frac{2}{5}$ | 1

Show your work

calculator · editing task choice

34. At 6 a.m. the temperature was -6°F in a walk-in freezer. It rose 20°F and then fell 6°F .

What is the temperature now, in $^{\circ}\text{F}$? Enter your answer.

Answer: _____

Show your work

calculator · equation editor

35. Write 0.75 as a percent. Enter your answer without the % symbol.

Answer: _____

Show your work

calculator · equation editor

36. A carry-on bag must weigh no more than 36 pounds.

Select **all** the values of x that make the situation true.

- ☐ 47
- ☐ 36
- ☐ 34
- ☐ 42
- ☐ 39
- ☐ 29

Select all that apply.

Show your work

multiselect

37. A data set is shown.

8, 12, 13, 17, 23, 23

Select **all** the statements that are true about the data set.

- ☐ The median is 17.
- ☐ The mean is 16.
- ☐ The median is 15.
- ☐ The mean is 19.
- ☐ The range is 15.
- ☐ The range is 19.

Select all that apply.

Show your work

multiselect

38. Select **all** the equations that have the solution $x = 6.1$.

- ☐ $x - 2 = 9.1$
- ☐ $x + 1.5 = 7.6$
- ☐ $x \div 2 = 6.1$
- ☐ $x + 1.5 = 6.1$
- ☐ $2x = 12.2$
- ☐ $x - 0.5 = 5.6$

Select all that apply.

Show your work

multiselect

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume